

LC76G Series

Reference Design

GNSS Module Series

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Status: Released



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About the Document

Document Information

Title	LC76G Series Reference Design
Subtitle	GNSS Module Series
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Document Status	Released

Revision History

Version	Date	Description
-	2022-05-11	Creation of the document
1.0	2022-07-14	First official release
1.1	2023-02-09	- Added the applicable variant LC76G (PB). - Reserved the AP_REQ pin and deleted the corresponding circuit. - Added I2C function for LC76G (PA). - Added the level-shifting circuits for UART interface (Sheets 1 and 3). - Updated VCC power supply control circuit (Sheet 2). - Updated the note about supported messages for UART (Sheet 4). - Deleted 1PPS indication circuit. - Added the optional notch circuit and band-pass filter circuit to active and passive antenna reference designs, as well as the corresponding note (Sheet 5). - Added the SCH and PCB design checklists.
1.2	2025-04-08	- Updated the power supply control circuit for VCC and V_BCKP, and deleted the rechargeable battery solution for V_BCKP (Sheet 2). - Deleted notch circuit and corresponding description for antenna reference design (Sheet 5 and SCH Design Checklist). - Updated the notch circuit and the band-pass circuit to a SAW filter circuit (Sheet 5).

Version	Date	Description
		4. Added routing requirement for 1PPS, deleted the routing width requirement for V_BCKP, and updated the routing width requirement for VCC (PCB Design Checklist).

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1 Reference Design

1.1. Introduction

This document provides the reference design of Quectel LC76G GNSS module, including the design of block diagram, MCU and power supply circuits, UART level-shifting circuit, module interfaces and antenna interface. It also comprises SCH and PCB design checklists.

The LC76G series comprises three variants: LC76G (AB), LC76G (PA) and LC76G (PB).

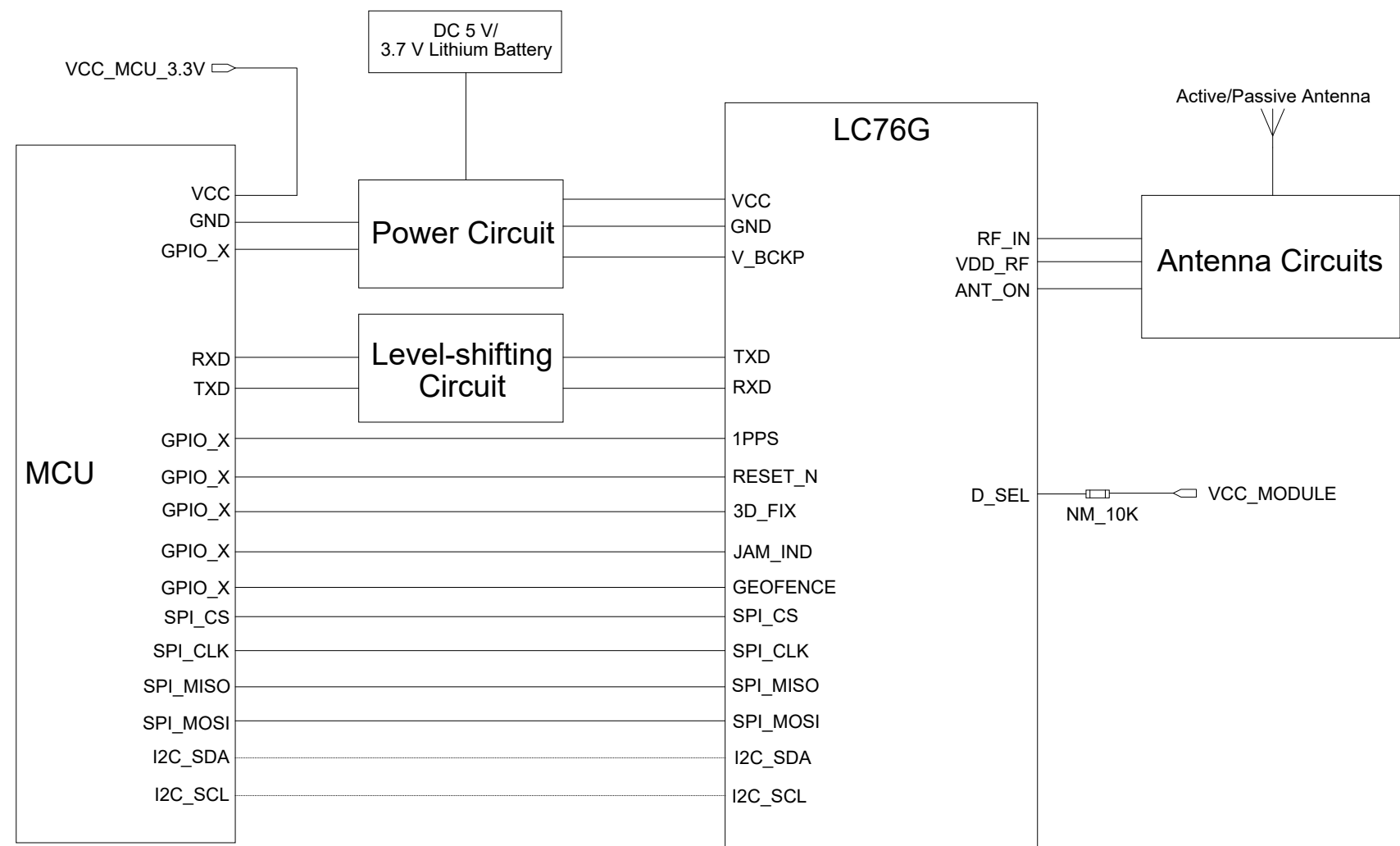
1.2. Reference Schematics and Design Checklists

The schematics and design checklists illustrated in the following pages are provided for your reference only.

NOTE

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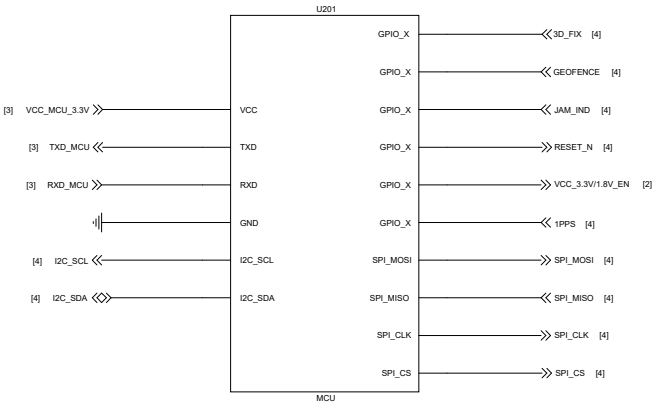
Block Diagram



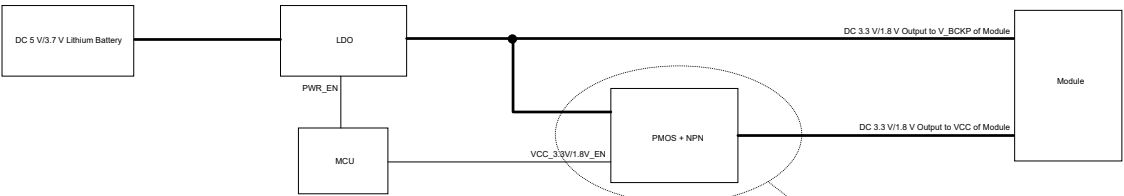
NOTE:
1. The D_SEL pin is internally pulled down by default and the UART is selected for firmware upgrade (in Download mode).
2. If SPI is selected for firmware upgrade (in Download mode), pull the D_SEL pin up to VCC externally with a 10 kΩ resistor.

MCU and Power Supply Circuits

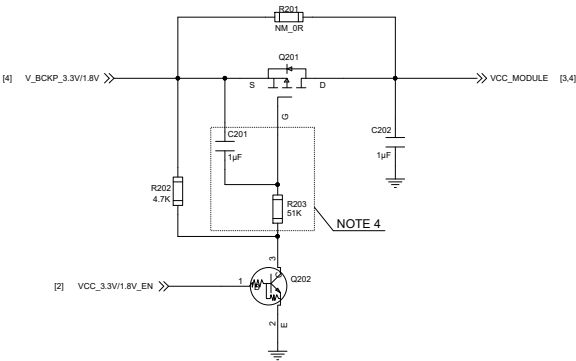
MCU Circuit



Power Supply Circuit



VCC Power Supply Control Circuit

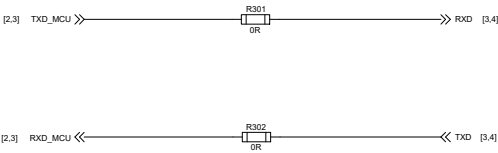


NOTE:

1. Ensure the module power supply is controlled by MCU to save power or restart the module when the module enters abnormal state.
2. The V_BCKP should always be powered if hot (warm) start is needed.
3. Select the appropriate lithium battery capacity according to the power consumption of the GNSS module.
4. C201 and R203 form a soft start circuit. The capacitance (C201) and resistance (R203) can be adjusted to set the soft start time according to your requirement.

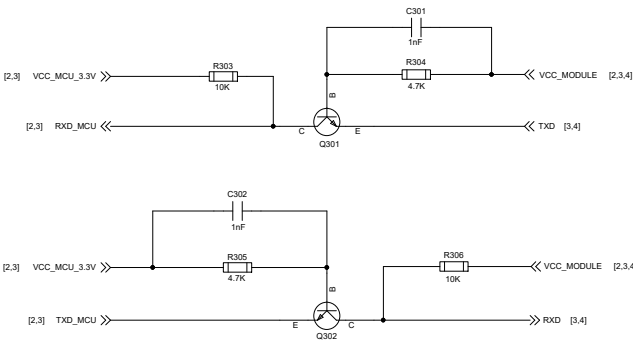
UART Level-shifting Circuits

LC76G (AB, PA) — Resistor Circuit Solution



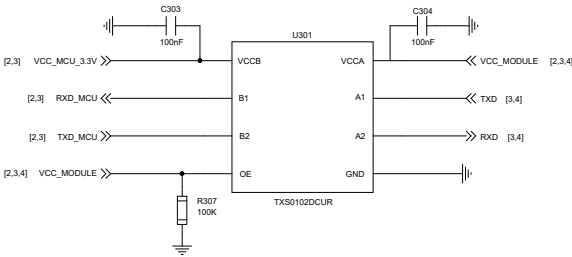
NOTE:
1. R301 and R302 are reserved for debugging the waveform of UART.
2. The above circuit is designed when UART voltage of MCU is consistent with that of the module.

LC76G (PB) — Transistor Solution



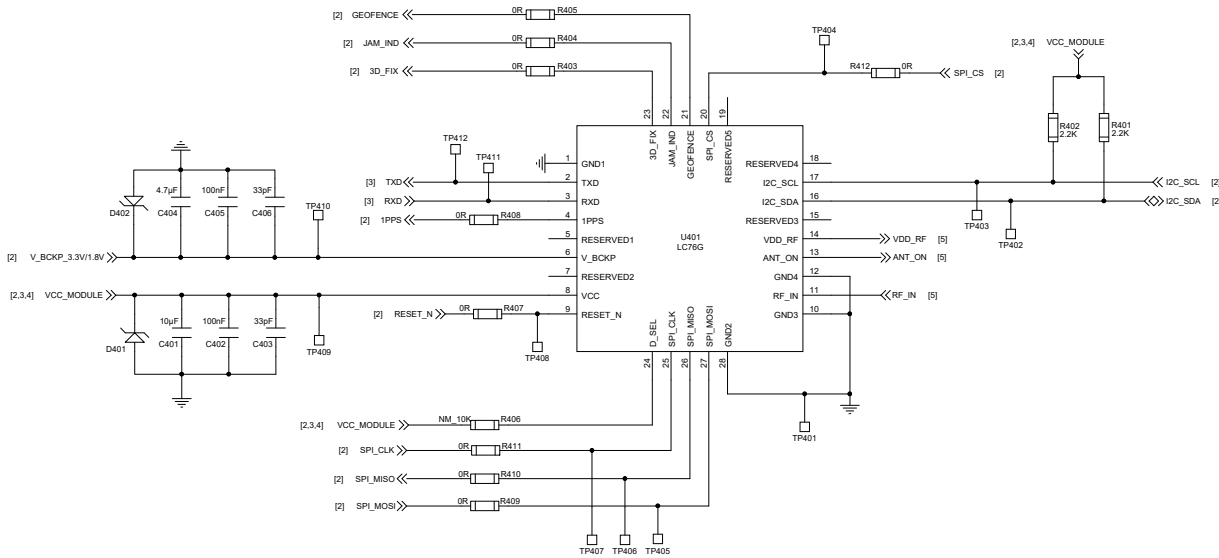
NOTE:
The above circuit is designed to achieve the level shifting between VCC_MCU_3.3V and VCC_MODULE, and block the leakage current from a power-on device to a power-off device.

LC76G (PB) — IC Solution



NOTE:
1. The IC solution generally requires VCCA ≤ VCCB. Please pay attention to the voltage relationship when using it.
2. The above circuit is designed to achieve the level shifting between VCC_MCU_3.3V and VCC_MODULE.

Module Interfaces

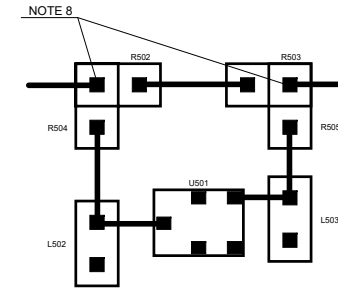
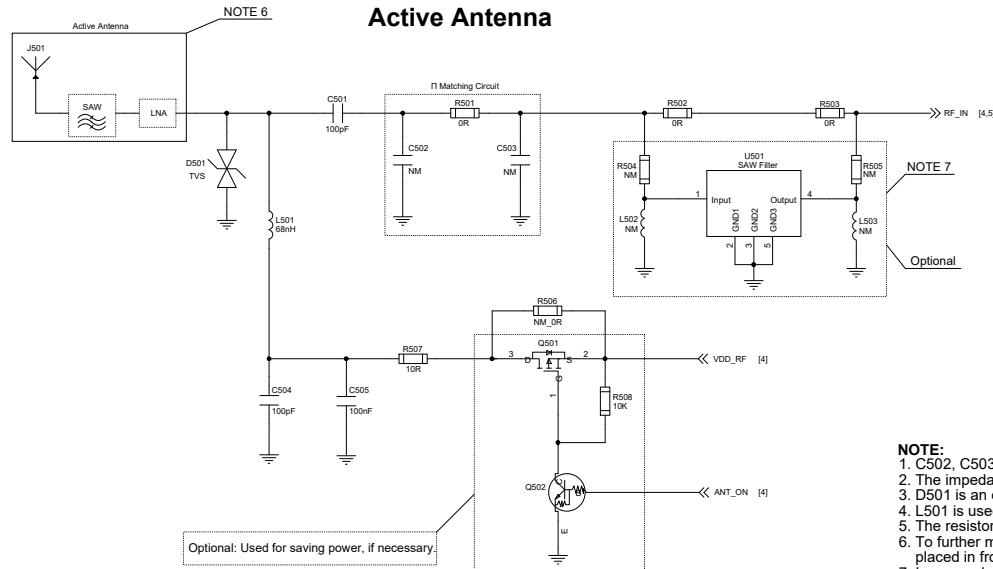


NOTE:
1. The power supply design must meet the sequence requirements in hardware design. For more information, see the related hardware design document.
2. A level-shifting circuit must be used when the I/O voltage of MCU is not matched with that of the module.
3. Externally pull D_SEL up with a 10 kΩ resistor to select SPI for firmware upgrade (in Download mode).
4. The RESET_N pin is internally pulled up to 1.8 V; the pin must be connected so that it can be used to reset the module if the module enters an abnormal state.
5. D401 and D402 are surge and ESD protection components, which are recommended to be placed near the VCC and V_BCKP respectively.
6. Reserve test points for debugging the module.

Module	LC76G (AB)	LC76G (PA)	LC76G (PB)
Voltage			
VCC_MODULE	2.55-3.6 V	2.55-3.6 V	1.75-1.98 V
I/O Level	Following VCC	Following VCC	Following VCC
V_BCKP	1.65-3.6 V	1.65-3.6 V	1.65-3.6 V

Antenna Interface

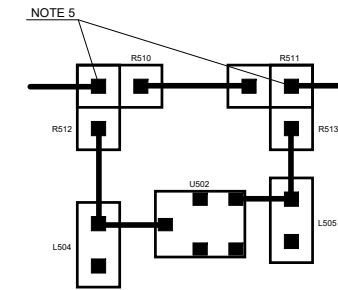
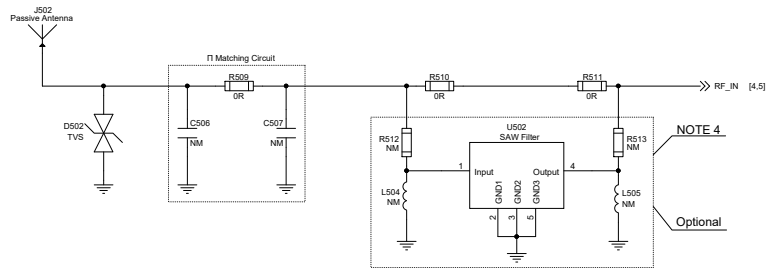
Active Antenna



NOTE:

1. C502, C503 and R501 form a Π matching circuit for antenna impedance modification. By default, R501 is 0 Ω , and C502 and C503 are not mounted.
2. The impedance of the RF trace line on the main PCB should be controlled to 50 Ω and the trace length should be kept as short as possible.
3. D501 is an electrostatic discharge (ESD) protection device to protect the RF signal input from the potential damage caused by ESD.
4. L501 is used for preventing the RF signal from leaking into the VDD_RF and preventing noise propagation from the VDD_RF to the antenna.
5. The resistor R507 is used for protecting the module in case the active antenna is short-circuited to the ground plane.
6. To further mitigate the impact of out-of-band signals on the GNSS module, you must choose the active antenna whose SAW filter is placed in front of the LNA in the internal framework. DO NOT place the LNA in the front.
7. In a complex electromagnetic environment, a SAW filter circuit must be added to further reduce the impact of out-of-band signals on the GNSS module. It should be placed close to the RF_IN pin during PCB design.
8. The resistors R502 and R504 need to share a pad, and R503 and R505 need to share a pad.
9. For more information, see the related hardware design document.

Passive Antenna



NOTE:

1. C506, C507 and R509 form a Π matching circuit for antenna impedance modification. By default, R509 is 0 Ω , and C506 and C507 are not mounted.
2. The impedance of the RF trace line on the main PCB should be controlled to 50 Ω and the trace length should be kept as short as possible.
3. D502 is an electrostatic discharge (ESD) protection device to protect the RF signal input from the potential damage caused by ESD.
4. In a complex electromagnetic environment, a SAW filter circuit must be added to further reduce the impact of out-of-band signals on the GNSS module. It should be placed close to the RF_IN pin during PCB design.
5. The resistors R510 and R512 need to share a pad, and R511 and R513 need to share a pad.
6. For more information, see the related hardware design document.

Quectel Wireless Solutions

PROJECT	LC70G Series	VER	1.2
DRAWN BY	Vijay ZHAO/Damen DONG	CHECKED BY	Storm BAO
DATE	2025-04-08	SHEET	5 OF 5

Table 1: SCH Design Checklist

Pin No.	Pin Name	Checklist	Result			Comment
			Pass	Fail	N/A	
1	GND	Reference ground of the module. The GND pin must be connected to ground.				
2	TXD	Connect to MCU through 0 Ω resistors or a level-shifting circuit. Reserve test points.				
3	RXD					
4	1PPS	Connect to the GPIO of the MCU.				
5	RESERVED	The RESERVED pin must be left N/C (not connected).				
6	V_BCKP	1. It is recommended to place a TVS, and a combination of a 4.7 μF, a 100 nF and a 33 pF decoupling capacitor near the V_BCKP pin. 2. Ensure that V_BCKP is controlled by MCU. 3. Reserve a test point. 4. V_BCKP must be connected to power supply for startup, and it should always be powered if hot (warm) start is needed.				
7	RESERVED	The RESERVED pin must be left N/C.				
8	VCC	1. It is recommended to place a TVS, and a combination of a 10 μF, a 100 nF and a 33 pF decoupling capacitor near the VCC pin. 2. Ensure that VCC is controlled by MCU. 3. Reserve a test point.				
9	RESET_N	Connect to the GPIO of the MCU through a 0 Ω resistor to control the module reset. Reserve a test point.				
10	GND	Reference ground of the module. The GND pin must be connected to ground.				

Pin No.	Pin Name	Checklist	Result			Comment
			Pass	Fail	N/A	
11	RF_IN	1. π matching circuit must be added for impedance modification. 2. In a complex electromagnetic environment, a SAW filter circuit must be added to reduce the impact of out-of-band signals interference. 3. It is recommended to select an ESD protection device with junction capacitance lower than 0.6 pF. 4. The inductor used in the power supply circuit of the active antenna is at least 68 nH and the inductor's pad which is close to RF_IN should be placed on the RF line.				
12	GND	Reference ground of the module. The GND pin must be connected to ground.				
13	ANT_ON	ANT_ON is connected to the transistor's base to control the power supply of VDD_RF for external active antenna or LNA.				
14	VDD_RF	Supplies power for an external active antenna or LNA.				
15	RESERVED	The RESERVED pin must be left N/C.				
16	I2C_SDA	1. The pins need to be pulled up externally to VCC with a 2.2 kΩ resistor respectively. 2. Connect them to MCU with a level-shifting circuit for LC76G (PB). 3. Reserve test points. 4. If unused, leave the pins N/C.				
17	I2C_SCL					
18	RESERVED	The RESERVED pin must be left N/C.				
19	RESERVED	The RESERVED pin must be left N/C.				
20	SPI_CS	Connect to MCU through a 0 Ω resistor. Reserve a test point.				
21	GEOFENCE	Connect to the GPIO of the MCU.				

Pin No.	Pin Name	Checklist	Result			Comment
			Pass	Fail	N/A	
22	JAM_IND	Connect to the GPIO of the MCU.				
23	3D_FIX	Connect to the GPIO of the MCU.				
24	D_SEL	Connect a 10 k Ω resistor in series to VCC. The resistor is not mounted by default.				
25	SPI_CLK	Connect to MCU through a 0 Ω resistor. Reserve a test point.				
26	SPI_MISO	Connect to MCU through a 0 Ω resistor. Reserve a test point.				
27	SPI_MOSI	Connect to MCU through a 0 Ω resistor. Reserve a test point.				
28	GND	Reference ground of the module. The GND pin must be connected to ground.				

NOTE

1. All GND pins must be connected to ground and reserved a GND test point. Leave RESERVED and unused pins N/C (not connected).
2. Quectel also provides design review services. It is strongly recommended that you submit your schematics and PCB designs to Quectel Technical Support for a formal review.

Table 2: PCB Design Checklist

Pin No.	Pin Name	Checklist	Result			Comment
			Pass	Fail	N/A	
1	GND	1. Confirm that there are no isolated shapes in the ground layer. 2. Module GND pads must be completely covered by the ground plane.				
2	TXD	1. Surround the signal traces with ground. 2. Keep the routing short and away from interference.				
3	RXD					
4	1PPS	1. Surround the signal trace with ground. 2. Avoid routing near strong interference. 3. Keep the routing as short as possible and keep fewer vias to avoid parasitic capacitance and inductance.				
5	RESERVED	/				
6	V_BCKP	1. The power supply first passes through the TVS, and then through the subsequent components. 2. The capacitors are placed near the power supply pin in descending order of capacitance. At least one GND via must be placed near the grounded end of the capacitor. If needed, there should be more than one GND via to meet the requirements. 3. The power routing and sensitive signal routings (with Clock, USB, MIPI, RF, etc.) must be isolated.				
7	RESERVED	/				
8	VCC	1. The power supply first passes through the TVS, and then through the subsequent components. 2. The capacitors are placed near the power supply pin in descending order of capacitance. At least one GND via must be placed near the grounded end of the capacitor. If needed, there should be more than one GND via to meet the requirements.				

Pin No.	Pin Name	Checklist	Result			Comment
			Pass	Fail	N/A	
		3. Ensure the power supply routing width is sufficient. The longer the routing, the wider it should be. The power routing and sensitive signal routings (with Clock, USB, MIPI, RF, etc.) must be isolated.				
9	RESET_N	Surround the signal trace with ground, and avoid routing near the strong interference.				
10	GND	1. Confirm that there are no isolated shapes in the ground layer. 2. Module GND pads must be completely covered by the ground plane.				
11	RF_IN	1. The characteristic impedance of the RF signal line(s) is kept at 50 Ω , and the RF trace is as short and straight as possible, with smooth lines (without bumps, with consistent geometry—it would be ideal for the footprints to be blended into the RF trace, with curved rather than sharp angles). 2. Ensure that there are no vias in the RF signal path. 3. Ensure that RF signal path is surrounded by ground. 4. RF signal line(s) and GNSS antenna are kept away from noise sources such as MCU(s), crystal(s) and other RF antenna(s).				
12	GND	1. Confirm that there are no isolated shapes in the ground layer. 2. Module GND pads must be completely covered by the ground plane.				
13	ANT_ON	1. Surround the signal trace with ground. 2. Avoid routing near strong interference.				
14	VDD_RF	Power routing should be surrounded by GND and avoid being parallel with other line(s).				
15	RESERVED	/				
16	I2C_SDA	1. Surround the signal traces with ground. 2. Keep the routing short and stay away from interference.				
17	I2C_SCL					

Pin No.	Pin Name	Checklist	Result			Comment
			Pass	Fail	N/A	
18	RESERVED	/				
19	RESERVED	/				
20	SPI_CS	1. Surround the signal trace with ground. 2. Keep the routing short and stay away from interference.				
21	GEOFENCE	1. Surround the signal trace with ground. 2. Avoid routing near strong interference.				
22	JAM_IND	1. Surround the signal trace with ground. 2. Avoid routing near strong interference.				
23	3D_FIX	1. Surround the signal trace with ground. 2. Avoid routing near strong interference.				
24	D_SEL	1. Surround the signal traces with ground. 2. Place the pull-up resistor close to the pin.				
25	SPI_CLK	1. Surround the signal trace with ground. 2. Keep the routing short and stay away from interference.				
26	SPI_MISO	1. Surround the signal trace with ground. 2. Keep the routing short and stay away from interference.				
27	SPI_MOSI	1. Surround the signal trace with ground. 2. Keep the routing short and stay away from interference.				
28	GND	1. Confirm that there are no isolated shapes in the ground layer. 2. Module GND pads must be completely covered by the ground plane.				

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